Egg Segmentation and Tracking

Computer vision internship assignment

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Code and results: https://github.com/sarangp17/egg-segmentation-tracking

Annotated tracking video (public Google Drive):
https://drive.google.com/file/d/1nzwb-UZDESOID0dqhEN9Ej4_-i6Tr7pG/view?usp=sharing

Summary

I built a pipeline in four stages: (1) pseudo-ground-truth egg masks for all 110 dataset images from the pretrained Segment Anything Model (SAM) plus a selection rule and manual box prompts, (2) a leakage-aware 80:20 split and a classical OpenCV segmentation developed through six logged experiments, (3) an evaluation of the classical masks against the SAM masks on the held-out images, and (4) segmentation and tracking of the eggs in a supplied video.

• Reference masks: 93 of 110 images were produced automatically, 17 of 110 (15.5 percent) needed manual box prompts to SAM.

• Classical method (final, E6): macro IoU 0.496 on the 88 development images and 0.409 on the 22 held-out evaluation images.

• Video: 744 frames (10 s to 41 s) processed. The tracker created 8 IDs for 2 real eggs; IDs are lost when a hand lifts or hides an egg.

Important: all scores measure agreement with SAM-based pseudo-ground truth, not accuracy against human labels. Tables below are generated from the result files saved by the notebook; the narrative numbers are copied from the notebook outputs.

1. Data

• Roboflow Universe dataset fire-bf7qe/egg-vzhtr, version 1, licence CC BY 4.0, 110 JPG images at 640x640 (Roboflow resized them with stretch).

• I exported it in the 'folder' format, which contains images only (no boxes or masks), so SAM masks are the only reference available. Two images sit in an 'Unlabeled' folder; both are boiling eggs in water.

• Roboflow's own train/valid/test folders (77/22/11 images) were ignored. All 110 images were pooled and re-split by me.

• Eggs are white. Backgrounds vary (tray, table cloth, wall, foil, grass, water); a few images have two touching eggs; there are shadows, and the two boiling eggs are translucent.

2. Task 1: pretrained segmentation (SAM)

Model: SAM ViT-B (segment-anything) on a Colab T4 GPU, automatic mask generator (32 points per side, predicted-IoU threshold 0.88, stability threshold 0.92, minimum region 200 pixels).

SAM returns many regions per image (eggs, tray, cloth, background), so I did not treat all proposals as eggs. A region is kept as an egg only if it passes simple checks, then the kept regions are merged and holes are filled:

• size between 2 and 90 percent of the image; solidity (area / convex-hull area) at least 0.88; aspect ratio at most 2.5;

• covers at most 20 percent of the image border (backgrounds wrap around the border, eggs sit inside); mean saturation at most 70; mean brightness at least 170.

• If an image has no passing region, the brightness limit is relaxed to 155 (used for 5 images).

The rule was developed iteratively by looking at overlays. Early versions failed in instructive ways: a rule that dropped regions larger than 45 percent of the image removed the eggs in close-up photos (the egg was region 0); a brightness limit of 155 for every image added false regions (a white panel, a tile, a plastic object); and one-step rules could not separate the translucent boiling eggs from the water.

Manual work (disclosed): 17 of 110 images (15.5 percent) used manual box prompts that I drew after inspecting the images: 12 images use boxes only (boiling eggs, eggs on foil, in grass, held between fingers, an egg on a tray with empty pockets, and others where colour rules gave nothing) and 5 images use the automatic rule plus an extra box for a missed egg. For each box, SAM's mask is cut to the box and only the largest connected piece is kept. Final reference blobs per image: 1 egg in 83 images, 2 in 16, 3 in 11.

3. Task 2: split and classical segmentation

3.1 Leakage-aware split

Each image got a 64-bit difference hash (dHash: shrink to 9x8 grey pixels and compare neighbouring pixels). Images whose hashes differ in at most 10 bits were grouped as near-duplicates (union-find), and whole groups were assigned to the evaluation set with a seeded shuffle (seed 42) until it held about 20 percent of the images.

• Result: 88 development and 22 evaluation images. Shared groups: 0. Shared images: 0.

• The smallest distance between any two images was 12 bits (median 21), so no near-duplicates existed at threshold 10; every image became its own group and the split reduced to a seeded random split. The closest development/evaluation pair differs by 16 bits and the two photos look different.

• Limits of perceptual hashing: it catches copies, resizes and small crops but can miss the same egg photographed from another angle or under other lighting; the threshold is a judgement call.

• Reference-mask methods per split: development 73 automatic, 2 automatic plus box, 11 box only, 2 relaxed-automatic; evaluation 15, 3, 1 and 3 respectively. Boxes were used for 13 of 88 development images and 4 of 22 evaluation images.

• The evaluation images were not used for choosing classical parameters. (The SAM selection rule was inspected on images from the whole dataset before the split, see limitations.)

3.2 Experiment log (development set only)

Trial	Change	Dev IoU	Decision	Observations
E1	Baseline: grayscale + global Otsu threshold, nothing else	0.345	keep as baseline, not final	Macro IoU 0.345, precision 0.38, recall 0.87. Finds eggs but also marks bright backgrounds and a finger; images 20 and 44 almost fully missed (egg not bright).
E2	Replace grayscale Otsu with HSV rule: saturation <= sat_max AND value >= val_min	0.4241	keep, current best	Macro IoU 0.424 (up from 0.345), precision 0.55, recall 0.70. Best sat_max=40, val_min=170. Dim or shadowed eggs (images 39, 20, 119, 44) almost fully missed; white panel behind egg (125) and a finger still marked.
E3	Add morphological opening then closing (elliptical kernel) after the E2 colour rule	0.4468	keep, current best	Macro IoU 0.447 (up from 0.424), precision 0.586, recall 0.690. Opening kernel 9 best (largest value tried, so larger not explored); closing hurt in every setting and was dropped. Dim-egg images and white panel image still fail.

Trial	Change	Dev IoU	Decision	Observations
E4	Add connected-component blob filter (min size, border contact, aspect ratio) after E3	0.4924	keep, current best	Macro IoU 0.492 (up from 0.447), precision 0.688, recall 0.660. Best min_area 2% of image; border-contact filter hurt in every setting (close-up eggs touch the border) so max_edge=1.0 (off). New zero-IoU images 149 and 150 suggest the filter removed their eggs.
E5	Insert distance-transform watershed (per-blob peaks) between E3 cleaning and the E4 blob filter	0.442	discard	Macro IoU 0.442 (worse than E4 0.492), recall dropped to 0.543. Best peak_frac was 0.5, the setting that splits least, suggesting watershed over-splits single eggs and the cut lines remove pixels (not verified).
E6	Keep E3 + E4 filter (min area 2%, no border rule); only raise max_aspect	0.4963	keep as FINAL algorithm	Macro IoU 0.496 (E4 was 0.492). max_aspect 3.0 recovered touching-egg blobs (images 149/150). Worst images are dim or shadowed eggs; blue in error maps is mostly egg pixels below the brightness limit.
V1	Video only: retune HSV saturation/brightness limits on 4 frames (oval proxy from manual boxes); opening and blob filter unchanged	0.3037	discard	Looser colour limits recovered some eggs but also marked shirts and pants; colour alone cannot separate eggs from pale or grey clothing.
V2	Video only: V1 colour sweep + tracking zone y=160-440 + max blob size 12% of frame; opening and blob filter as E6	0.4431	discard (table still marked)	Brightness limit is the key setting (proxy IoU 0.20 at 170 vs 0.44 at 120). Zone + size limit removed shirts/pants, but table still marked at 22-34 s (blobs with brightness 111-163, solidity 0.55-0.79) and one egg missed at 36 s.
V3	Video only: bright, low-saturation core -> opening -> small dilation -> keep only where loose rule (sat<=100, val>=120) agrees; zone y=160-440; blob filter min 1%, aspect 3.0	0.5538	keep, final video segmentation	Bright core (sat<=60, val>=170) + opening + 21 px grow, limited to loose rule (sat<=100, val>=120), zone y=160-440. Proxy IoU 0.554 on 4 boxed frames (V2 0.443); right egg count in 3 of 4 frames; 16-frame preview judged good by visual check.

E1 to E6 report macro IoU on the 88 development images against the SAM masks. V1 to V3 are the video-only trials (section 5); their score is a proxy IoU on 4 hand-boxed frames and is not comparable with E1 to E6. The full log with parameters and motivations is experiments/experiment_log.csv in the repository.

What the experiments showed:

• E1 (grayscale + Otsu) found most egg pixels but also bright backgrounds and fingers: recall 0.87, precision 0.38.

• E2 (HSV: saturation at most 40, brightness at least 170) removed much of the coloured background; precision rose to 0.55 but dim eggs were lost. A 3x3 sweep was used.

• E3 added morphological opening (kernel 9). Closing reduced the score in every setting (it merged eggs with bright background) and was dropped.

• E4 added a blob filter (minimum 2 percent of the image). The border-contact filter hurt in every setting because close-up eggs touch the border, so it was switched off. E4 also removed some real eggs: touching eggs form long blobs.

• E5 (distance-transform watershed to split touching eggs) lowered the score to 0.442 and was discarded. The best setting was the one that splits least, so over-splitting is the likely reason, but I did not verify it.

• E6 raised the maximum blob aspect ratio from 2.5 to 3.0 (a diagnosis of two images showed egg blobs with aspect 2.7 to 3.6 being removed), giving the final 0.496. Tuning stopped here to avoid overfitting the development set.

4. Task 3: evaluation against the pseudo-ground truth

The SAM masks are treated as reference and the classical masks as predictions. Per image I compute IoU (overlap divided by union), Dice, precision and recall on binary masks. Empty-mask rules are fixed: if both masks are empty the score is 1; precision of an empty prediction is 1 only when nothing was expected; recall is 1 when nothing was expected. **Macro** averages are the mean of per-image scores (every image

counts equally); **micro** averages pool the pixels of all images (large eggs and large false regions count more). The evaluation set was used once, after the algorithm was frozen.

Metric	Development (88)	Evaluation (22)
Macro IoU	0.496	0.409
Macro Dice	0.618	0.525
Macro precision	0.684	0.577
Macro recall	0.677	0.647
Median IoU	0.537	0.475
Micro IoU	0.400	0.280
Micro Dice	0.571	0.437
Micro precision	0.496	0.337
Micro recall	0.674	0.622

Egg counting on the evaluation set: the number of predicted blobs equals the reference count in 12 of 22 images (mean absolute error 0.64 eggs). The evaluation score is lower than the development score; with only 22 images and a large spread (IoU standard deviation 0.27) this gap is consistent with mild overfitting to the development set and with sampling noise, and should not be over-interpreted.

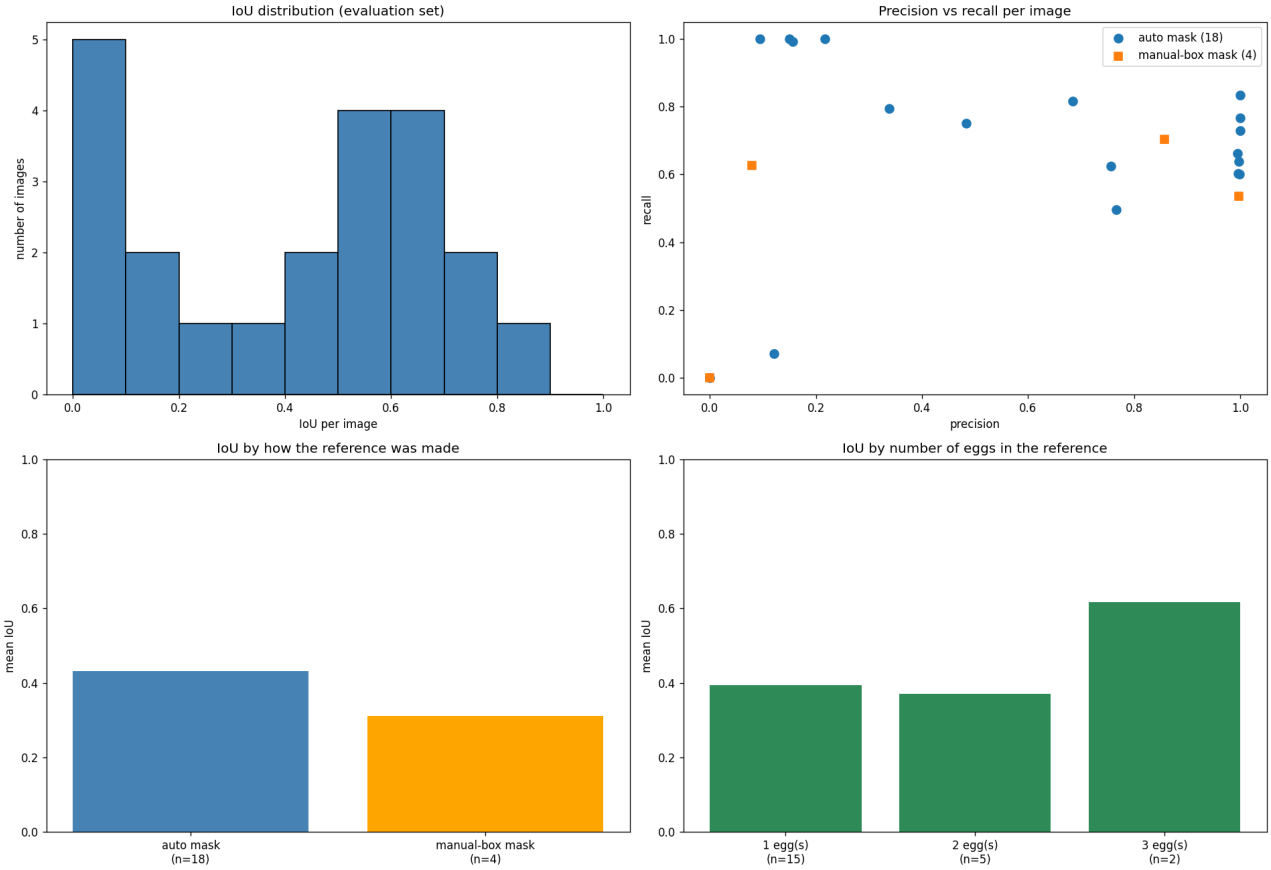


Figure 1: IoU distribution, precision against recall, IoU by reference type (automatic or box), and IoU by number of eggs (evaluation set).

Worst six evaluation images:

Image	Reference type	Ref eggs	Pred eggs	IoU	Precision	Recall
not_damaged_117_jpg.rf.d	auto_strict_plus_box	2	0	0.00	0.00	0.00
not_damaged_78_jpg.rf.d3	auto_relaxed	1	2	0.00	0.00	0.00

Image	Reference type	Ref eggs	Pred eggs	IoU	Precision	Recall
not_damaged_27_jpg.rf.9e	auto_relaxed	1	3	0.05	0.12	0.07
not_damaged_42_jpg.rf.66	manual_box	1	2	0.08	0.08	0.63
not_damaged_104_jpg.rf.3	auto_strict	1	1	0.09	0.09	1.00
not_damaged_111_jpg.rf.d	auto_strict	1	1	0.15	0.15	1.00

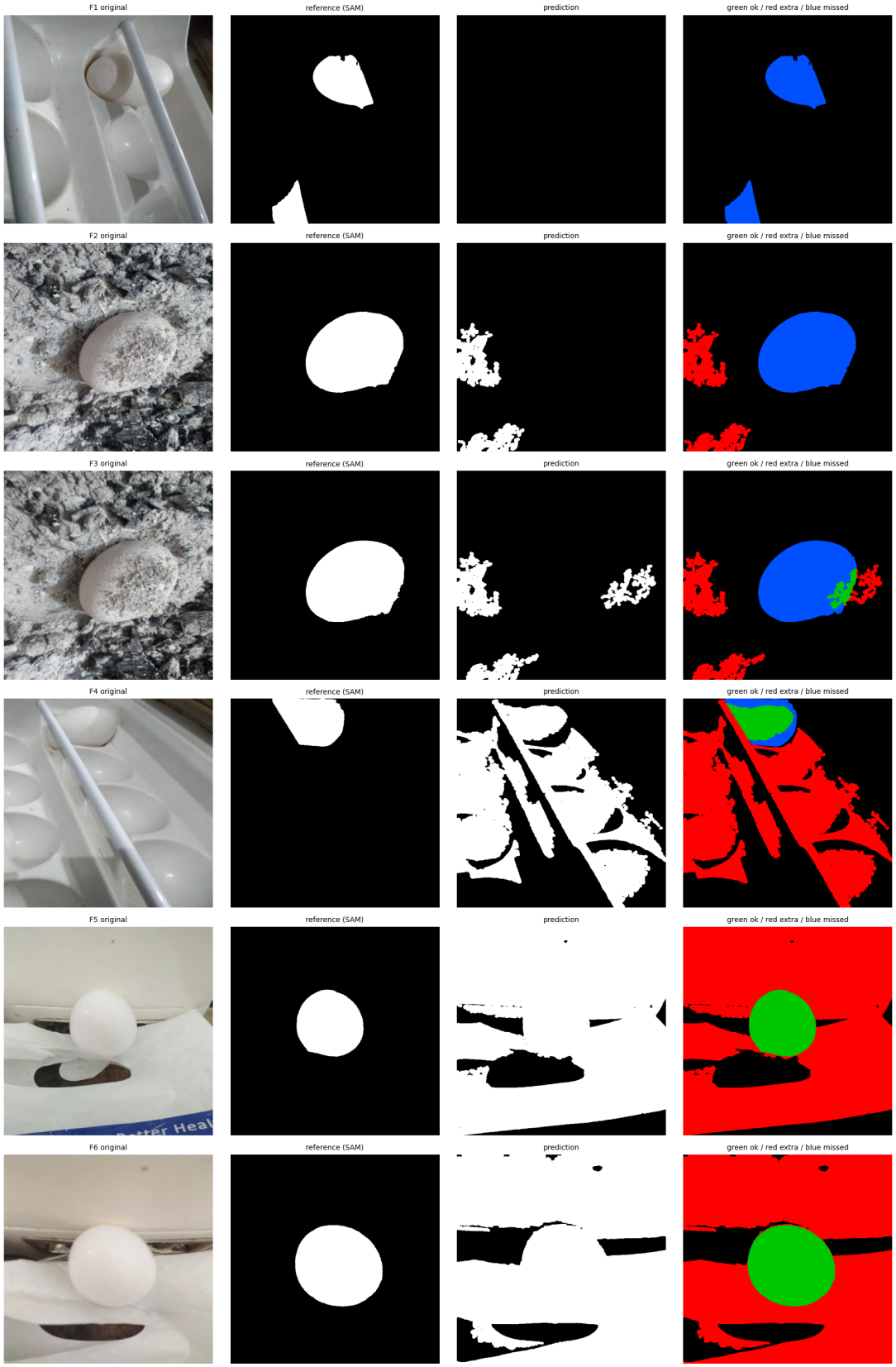


Figure 2: the six worst evaluation images (original, reference, prediction, error map: green correct, red false positive, blue missed).

Qualitative analysis

• Missed or partly missed eggs (blue): the brightness rule drops the shadowed side of an egg and dim or translucent eggs, which lowers recall to about 0.65. One evaluation image has IoU 0 with no egg predicted

at all.

• Large false positives (red): images with recall near 1 but precision around 0.1 mean the prediction covers the egg plus a much larger region. Colour alone cannot separate an egg from a white tray, panel or wall of the same colour; this appeared in development as well (a white panel behind an egg).

• Count errors: touching eggs can merge into one blob (under-segmentation) while textured backgrounds or tray pockets can add extra blobs (over-segmentation); 10 of 22 evaluation images have the wrong blob count.

• Shadows and reflections: closing and hole filling help only for small gaps; the dim side of an egg is the dominant shadow effect.

• Reference errors: in several development error maps the 'missed' (blue) pixels were fingers or tile patterns that SAM had marked as egg. In those cases the classical method is closer to the truth than the score suggests, so the true accuracy is not measured by these numbers.

5. Task 4: video segmentation and tracking

Video: 854x480, 24 FPS, 60.1 s, 1443 frames. Eggs are visible from about 10 s to 41 s (the rest is text slides), a hand moves and lifts the eggs, and the video contains 2 eggs in total. Only 10 s to 41 s (744 frames) was processed; every frame was used.

Segmentation. The dataset settings found only part of the eggs in the video (video eggs are darker). I tried three video-only versions, evaluated against ovals drawn inside boxes I marked on 4 frames (14, 20, 32 and 34 s) and checked visually on 16 preview frames:

• V1: looser colour limits recovered eggs but also marked shirts and trousers. Discarded.

• V2: V1 plus a tracking zone (image rows 160 to 440) and a maximum blob size removed the clothes; proxy IoU 0.443 but the table was still marked. Discarded.

• V3 (final): find a bright low-saturation core (saturation at most 60, brightness at least 170), apply opening, grow it by 21 pixels, and keep pixels only where a looser colour rule agrees, inside the zone. Proxy IoU 0.554; the right egg count in 3 of 4 boxed frames; judged good on the 16 preview frames.

These video settings (including the fixed tracking zone) were tuned on 4 frames and are specific to this video.

Tracking. For each blob I compute the centroid, bounding box and outline. Tracks and blobs are matched greedily by distance (closest pairs first, at most 100 px, plus 5 px per frame a track was unseen). A new track receives an ID after 3 frames (eggs visible in the first frame get one at once), a lost track is kept for 24 frames (1 s) so a returning egg keeps its ID, and a new blob within 40 px of an already matched blob is ignored to avoid duplicate IDs. The output video shows outlines, boxes, centroid trails and 'egg N' labels.

Results on the full run: tracked eggs per frame - 2 eggs in 506 frames, 1 egg in 231 frames, 0 in 7 frames. Eight IDs were created for the 2 real eggs:

ID	Frames seen	First (s)	Last (s)	Comment
1	225	10.0	19.3	first egg until it was lifted
2	494	10.0	31.0	long track; at the start the two close eggs formed one blob
3	3	21.0	21.1	egg lifted by the hand, then hidden
4	96	23.2	27.2	egg put back after more than 1 s hidden, new ID
5	1	28.3	28.3	one-frame noise
6	7	28.6	28.8	short noise near the hand
7	244	30.8	41.0	egg put back again, new ID
8	173	33.8	41.0	second egg re-detected after the hand moved it

Failure modes: an egg hidden or lifted for more than 1 second gets a new ID on return (identity is not preserved across hand occlusions); two close eggs can merge into a single blob and share one ID; tracks of one to seven frames are noise. A possible improvement is to keep lost tracks longer and ignore very short tracks; I did not have time to test it.

Annotated video (10 s to 41 s): https://drive.google.com/file/d/1nzwb-UZDESOID0dqhEN9Ej4_-i6Tr7pG/view?usp=sharing. I verified that the link plays in a private browser window without signing in.

6. Limitations

• The reference is SAM output, partly steered by my manual boxes (17 of 110 images). It is not human ground truth, and SAM's own errors count against the classical method. The scores measure agreement, not accuracy.

• The SAM selection rule was tuned by inspecting images from the whole dataset, including images that later fell into the evaluation set. The classical algorithm was tuned on the development set only.

• The evaluation set is small (22 images), so scores carry large uncertainty. The perceptual-hash grouping found no near-duplicates, so the split is effectively a seeded random split.

• A threshold-based method cannot handle dim, shadowed or translucent eggs, or backgrounds with the same colour as the eggs. Watershed did not help (over-splitting is a likely but unverified cause).

• Video settings were tuned on 4 frames with rough oval proxies and use a fixed zone; ID continuity breaks when a hand lifts or hides an egg.

• The src/ code in the repository is a cleaned copy of the notebook logic that was smoke-tested on synthetic data only; the notebook produced all reported numbers.

7. Setup and reproducibility

• Repository: https://github.com/sarangp17/egg-segmentation-tracking (README has setup commands, dataset download, notebook cell order, parameters in configs/final_params.json).

• Environment: Google Colab; SAM needs a GPU (T4 used), everything after the reference masks runs on CPU. Packages are listed in requirements.txt. SAM ViT-B checkpoint is downloaded separately (not in the repository).

• Dataset: download version 1 in 'folder' format with a (personal) Roboflow API key; images are not redistributed. The source video is downloaded separately.

• Seeds and parameters: split seed 42; every frozen parameter is in configs/final_params.json; every trial is in experiments/experiment_log.csv; the split is in experiments/split_manifest.csv.

8. AI-tool disclosure

I used Claude (Anthropic) throughout this project to plan the work, propose methods and parameter sweeps, write and debug code, and draft this report and the README. Claude wrote most of the code. My own work: I set up and ran every notebook in Google Colab; checked each stage visually (SAM overlays, error maps, video previews); supplied the manual box prompts for 17 of the 110 reference masks and the boxes used to tune the video settings; reported the failures I saw (for example clothes and the table marked as eggs, missed eggs), which led to changes in the method; and reviewed and approved the keep/discard decision for each experiment. Every number in this report comes from runs I executed. SAM is a pretrained model by Meta AI (Apache-2.0).

Credits: dataset 'egg' by a Roboflow Universe user (fire-bf7qe), CC BY 4.0; SAM by Meta AI (Apache-2.0); source video on YouTube (not redistributed).